

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2,3,4,!,2,5,I,2,3,4,5(No of Frames=3).

CODE:

```
GNU nano 6.2 lru.c *
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    int n = 12;

    int frame[10], time[10];
    int i, j, k, pos, min;
    int hits = 0, faults = 0, counter = 0;

    // Initialize frames
    for(i = 0; i < frames; i++) {
        frame[i] = -1;
    }

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check if page is already in frame (HIT)
        for(j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        // If page not found (FAULT)
        if(found == 0) {
            faults++;

            // Find least recently used page
            min = time[0];
            pos = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }
        }
    }

    printf("Hits: %d, Faults: %d, Counter: %d\n", hits, faults, counter);
    printf("Percentage Hit: %.2f%%\n", (float)hits / (float)counter * 100);
    printf("Percentage Fault: %.2f%%\n", (float)faults / (float)counter * 100);
}
```

```
GNU nano 6.2 lru.c *
for(j = 0; j < frames; j++) {
    if(frame[j] == pages[i]) {
        hits++;
        counter++;
        time[j] = counter;
        found = 1;
        break;
    }
}

// If page not found (FAULT)
if(found == 0) {
    faults++;

    // Find least recently used page
    min = time[0];
    pos = 0;

    for(j = 1; j < frames; j++) {
        if(time[j] < min) {
            min = time[j];
            pos = j;
        }
    }

    frame[pos] = pages[i];
    counter++;
    time[pos] = counter;
}

printf("\nTotal Page Hits = %d", hits);
printf("\nTotal Page Faults = %d", faults);

float hit_ratio = (float)hits / n * 100;
float fault_ratio = (float)faults / n * 100;

printf("\nHit Percentage = %.2f%%", hit_ratio);
printf("\nFault Percentage = %.2f%%\n", fault_ratio);

return 0;
}
```

Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark M-] To Bracket
Exit Read File Replace Paste Justify Go To Line M-E Redo M-6 Copy M-Q Where Was

500312 +4.85% Search 46 99+ ENG IN 2:06 PM 3/27/2026

OUTPUT:

```
who@Waghmare08: ~$ LRU.c
LRU.c: command not found
who@Waghmare08: ~$ nano lru.c
who@Waghmare08: ~$ gcc lru.c -o lru
who@Waghmare08: ~$ ./lru.c
-bash: ./lru.c: Permission denied
who@Waghmare08: ~$ ./lru

Total Page Hits = 2
Total Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%
who@Waghmare08: ~$ |
```